

# SCALA2

The fully integrated, self-priming pressure booster with intelligent constant pressure control. Specified here in full: curves, dimensions, materials and approvals.

**45 m**

MAXIMUM HEAD

**3 m<sup>3</sup>/h**

NOMINAL FLOW

**44 dB(A)**

TYPICAL NOISE

**10 bar**

MAX SYSTEM PRESSURE

**Possibility in every drop**

Performance data to ISO 9906 grade 2B

[grundfos.com](http://grundfos.com)



# One unit, fully integrated

SCALA2 combines pump, motor, pressure tank, sensor, drive and non-return valve in one compact enclosure. An integrated sensor measures discharge pressure continuously and the inverter adjusts speed to hold the setpoint, giving constant pressure regardless of inlet conditions or draw-off.



Serviceable by design

## Motor

Water-cooled permanent magnet motor with integrated frequency converter. Built-in thermal protection, no external motor protection required.

## Control

Default setpoint 3 bar (43 psi), adjustable on the operating panel. Soft starts limit starting and operating currents.

## Acoustics

Typical operation at 44 dB(A), measured at  $Q = 1 \text{ m}^3/\text{h}$  and  $H = 19 \text{ m}$  to ISO 3745.

## Pumped liquids

Clean, thin, fresh water. pH 4 to 9, chloride max 300 ppm, free chloride max 1.0 ppm.

## Protection functions

### Dry-running

Stops the pump without liquid.  
Always on, cannot be disabled.

### Micro-leakage

Alarms on frequent start-stop cycling caused by minor leaks.  
Default off.

### Max runtime

Stops the pump after 30 minutes continuous running. Default on.

## FEATURE SET

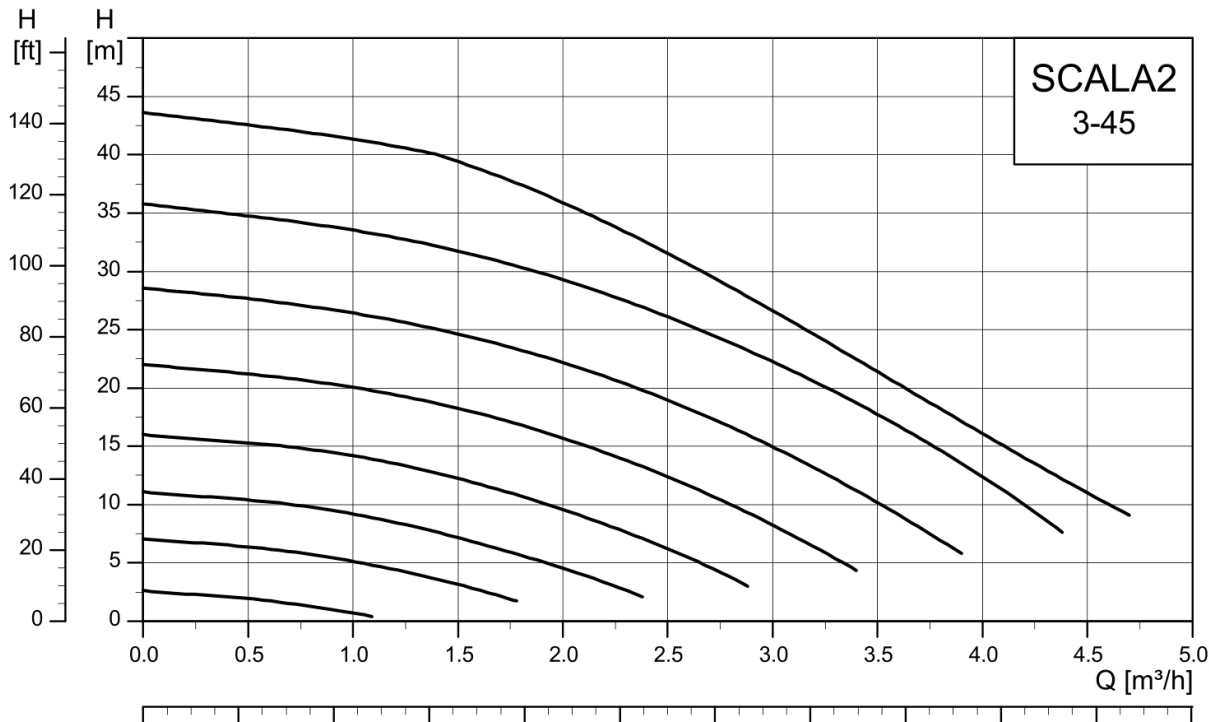
Auto start-stop . Self-priming . Soft start . Adjustable constant pressure . Integrated frequency converter . Alarm indication . Anti-cycling



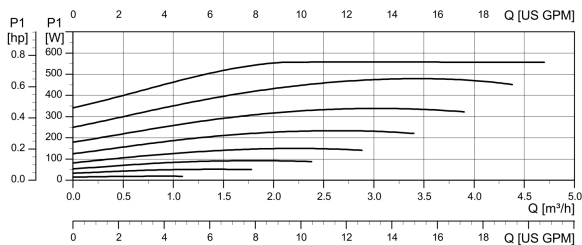
# Performance curves

Head, power input and suction performance for SCALA2 3-45, to ISO 9906 grade 2B.

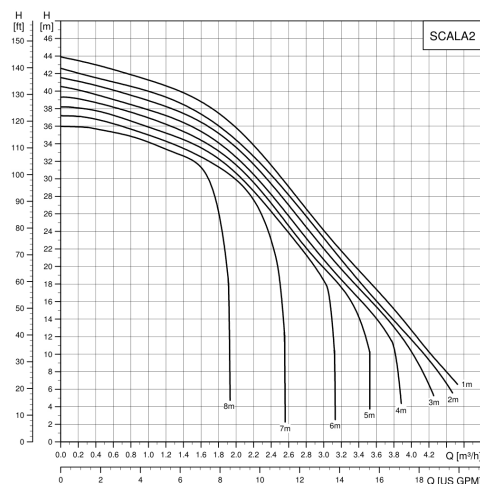
## Head vs flow . 50/60 Hz



## Power input P1



## Suction performance



Self-priming lift up to 8 m. Suction curves show available head at each suction depth. With positive inlet pressure, fit the optional locking pin to remove the priming clicking noise.

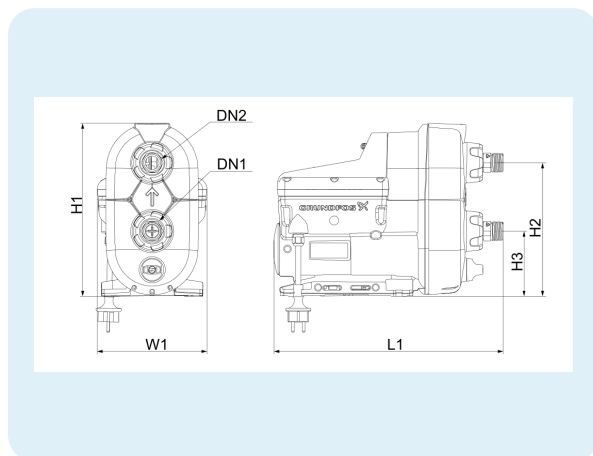
# Data and dimensions

## Operating conditions

|                                    |                  |
|------------------------------------|------------------|
| Max ambient temperature, 200-240 V | 55 °C (131 °F)   |
| Max ambient temp, 115/230 V 60 Hz  | 45 °C (113 °F)   |
| Max liquid temperature             | 45 °C (113 °F)   |
| Max system pressure                | 10 bar (145 psi) |
| Max inlet pressure                 | 6 bar (87 psi)   |
| Maximum head                       | 45 m (147 ft)    |
| Enclosure class                    | IP X4D, outdoor  |
| Noise level, typical               | < 44 dB(A)       |
| pH range                           | 4 to 9           |
| Chloride / free chloride, max      | 300 / 1.0 ppm    |

## Electrical data

| Supply        | Hz    | I <sub>max</sub> | P <sub>1</sub> | Standby |
|---------------|-------|------------------|----------------|---------|
| 1 x 200-240 V | 50/60 | 2.8 A            | 550 W          | 2 W     |
| 1 x 208-230 V | 60    | 2.8 A            | 550 W          | 2 W     |
| 1 x 115 V     | 60    | 4.9 A            | 550 W          | 2 W     |



## Dimensions

|                  |                  |
|------------------|------------------|
| H1 height        | 302 mm / 11.9 in |
| H2 outlet height | 234 mm / 9.2 in  |
| H3 inlet height  | 114 mm / 4.5 in  |
| L1 length        | 403 mm / 15.9 in |
| W1 width         | 193 mm / 7.6 in  |
| DN1 / DN2        | R 1 or NPT 1"    |
| Weight           | 10 kg / 22 lb    |

### LIQUID REQUIREMENTS

**pH 4 to 9**

**Chloride max 300 ppm**

**Free chloride max 1.0 ppm**

**8 m**

Self-priming suction lift. SCALA2 lifts water from below inlet level and handles a mix of air and liquid until fully primed, suited to tanks, shallow wells and rainwater systems.

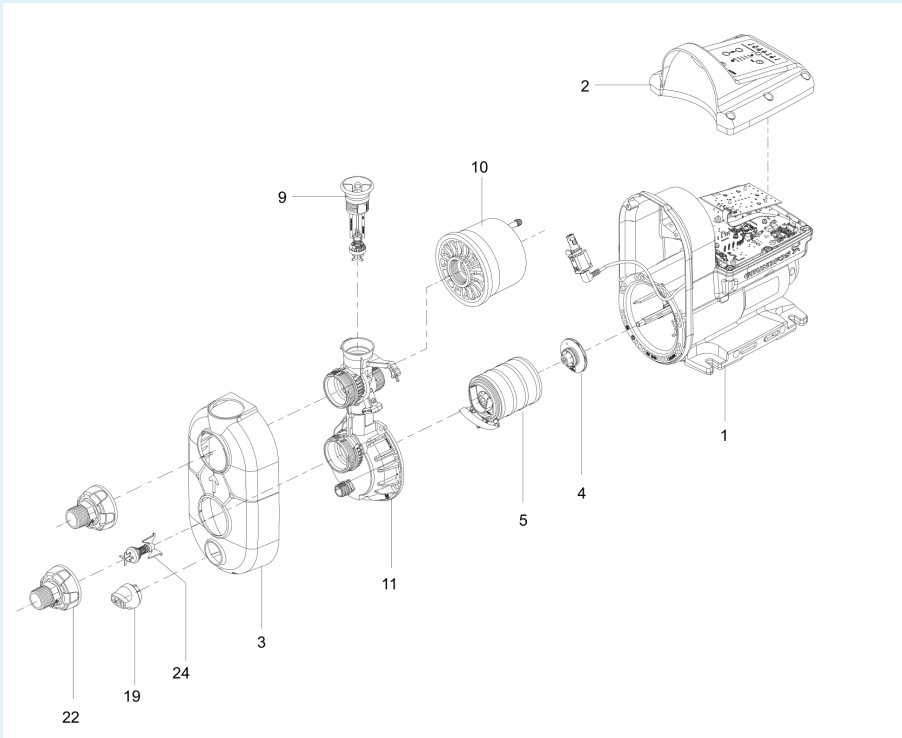
## Approvals and markings

Safety: CE, VDE, cULus, UKCA, EAC, PSE JQA, RCM, KC, SNI, S-mark. Drinking water: UL Classified to NSF/ANSI 61 and 372, WRAS, ACS, plus UBA and AWQC. Plug variants cover Schuko, UK, AUS, ARG, Chile, Thailand, India, Japan, NEMA 6-15P and NEMA 5-15P, and no-plug versions for fixed wiring.



# Construction and materials

Aluminium pump and stator housing, corrosion-resistant composite internals and a four-stage chamber stack. Wear parts marked below are available as service parts.



## Material specification

| Pos | Component                  | Material              |
|-----|----------------------------|-----------------------|
| 1   | Pump and stator housing    | Alu / comp / AISI 304 |
| 2   | Control box cover          | Composite             |
| 3   | Pump cover *               | Composite             |
| 4   | Shaft seal                 | Carbon / Ceramic      |
| 5   | Chamber stack, 4 stages    | Composite             |
| 9   | Non-return valve, outlet * | Composite             |

| Pos                           | Component                 | Material                |
|-------------------------------|---------------------------|-------------------------|
| 10                            | Pressure tank, complete   | Comp / Butyl / AISI 304 |
| 11                            | Connection part           | Composite               |
| 19                            | Drain plug *              | Composite               |
| 22                            | Thread R 1" / NPT 1" *    | Composite               |
| 24                            | Non-return valve, inlet * | Composite               |
| * Available as a service part |                           |                         |



# From box to running water

- 01 Connect

Connections adjust up to 5° for easy pipe alignment.
- 02 Prime

Fill once through the priming port. Self-priming from there.
- 03 Plug in

Standard outlet. Default constant pressure 3 bar.
- 04 Set precharge

Match tank precharge to your chosen setpoint.

## System notes

Maximum inlet pressure 6 bar; design the system for 10 bar. With water heaters, fit a non-return valve, pressure-relief valve or thermal expansion tank between SCALA2 and the heater. Mount on a solid foundation within 5 degrees of level, indoors or outdoors, with 0.5 m clearance on three sides. For suction lift above 4 m or inlet runs over 10 m, use inlet pipe larger than 1 inch.

## Accessories and product numbers

| Accessory                      | Connection | Product no. |
|--------------------------------|------------|-------------|
| Flexible hose set              | Straight   | 99891639    |
| Flexible hose set, brass elbow | Elbow      | 99891647    |
| Inlet filter, 250 micron       | 1" Rp      | 99725183    |
| Inlet filter, 250 micron       | 1" NPT     | 99725185    |
| Twin accessory set             | 1" Rp      | 99743076    |
| Twin accessory set             | 1" NPT     | 99743075    |
| Locking pin, 20 pieces         | All        | 99862078    |



Possibility in every drop

Size, select and download the full data booklet, CAD and BIM files in the Grundfos Product Center at [product-selection.grundfos.com](https://product-selection.grundfos.com)